Practical-6 : Study and implement various operations on Python strings such as creation, indexing, slicing, searching, modification through methods, splitting, joining, and immutability.

Use a Python string message to represent a text message and perform the following tasks:

1. Initialize the string message with a meaningful sentence.

```
message = "Hello World 123"
```

2. Display the complete string.

```
print(message)
```

```
Hello World 123
```

3. Display the first and last characters of the string.

```
print(message[0])
print(message[-1])
```

```
H
3
```

4. Display a substring from index 2 to index 10 using slicing.

```
print(message[2:10])
```

```
llo Worl
```

5. Find and display the length of the string using len().

```
print(len(message))
```

```
15
```

6. Convert the string to uppercase.

```
print(message.upper())
```

```
HELLO WORLD 123
```

7. Convert the string to lowercase.

```
print(message.lower())
```

```
hello world 123
```

8. Convert the string to title case.

```
print(message.title())
```

```
Hello World 123
```

9. Count the number of occurrences of a specific character or word.

```
print(message.count("l"))
```

```
3
```

10. Search for a given substring and display its starting index.

```
print(message.find("World"))
```

```
6
```

11. Check whether the string starts with a given substring.

```
print(message.startswith("Hello"))
```

```
True
```

12. Check whether the string ends with a given substring.

```
print(message.endswith("World"))
```

```
False
```

13. Replace a word in the string with another word.

```
print(message.replace("World", "Python"))
```

```
Hello Python 123
```

14. Split the string into a list of words.

```
print(message.split(" "))
```

```
['Hello', 'World', '123']
```

15. Join the list of words back into a single string using a separator.

```
print(" ".join(message.split(" ")))
```

```
Hello World 123
```

16. Remove leading and trailing whitespace using strip().

```
print(message.strip(" "))
```

```
Hello World 123
```

17. Find and display the number of vowels in the string.

```
vowels = "aeiouAEIOU"
vowel_count = sum(1 for char in message if char in vowels)
print("Number of vowels:", vowel_count)
```

```
Number of vowels: 3
```

18. Reverse the string using slicing.

```
print(message[::-1])
```

```
321 dlroW olleH
```

19. Check whether the string is a palindrome.

```
print(message == message[::-1])
```

```
False
```

20. Compare two strings and display whether they are equal.

```
print(message == "Hello World")
```

```
False
```

21. Concatenate two strings and display the result.

```
print(message + " " + "Python")
```

```
Hello World 123 Python
```

22. Demonstrate string immutability by attempting to modify a character.

```
try:
    message[0] = "h"
except TypeError as e:
```

```
    print("Error:", e)

modified_string = "h" + message[:]
print("Modified String:", modified_string)
```

```
Error: 'str' object does not support item assignment
Modified String: hHello World 123
```

23. Extract and display only the digits present in the string.

```
digits = ""
for char in message:
    if char.isdigit():
        digits += char
print("Digits in String:", digits)
```

```
Digits in String: 123
```

24. Count the number of words in the string.

```
print(message.count(" "))
```

```
2
```

25. Convert the string into a list of characters.

```
print(list(message))
```

```
['H', 'e', 'l', 'l', 'o', ' ', 'W', 'o', 'r', 'l', 'd', ' ', '1', '2', '3']
```